

1. Sejam dois primos $p=17$ e $q=41$ usados em um esquema RSA. Qual dos parâmetros $e_1=32$ ou $e_2=49$ é um expoente válido? Calcule a chave privada correspondente.

Step 1: Set prime numbers

As a starting point for RSA, choose two prime numbers p and q :

$p =$	17
$q =$	41

For the algorithm to work, the two prime numbers must be different.

► Safety note

e (Step 2.3)

The number e can now be chosen freely.
However, it must be **coprime** to $\phi(n)$.

$e =$	32
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❗ e is not coprime to $\phi(n)$

e (Step 2.3)

The number e can now be chosen freely.
However, it must be **coprime** to $\phi(n)$.

$e =$	49
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2. Faça a criptografia e descriptografia usando o algoritmo RSA (podem usar cryptool.org):

- $p=3$, $q=11$, $e=7$, mensagem=5

Step 1: Set prime numbers

As a starting point for RSA, choose two prime numbers p and q .

$p =$ 3

$q =$ 11

For the algorithm to work, the two prime numbers must be coprime.

e (Step 2.3)

The number e can now be chosen freely.

However, it must be **coprime** to $\phi(n)$.

$e =$ 7

RSA encrypts only numbers

Example:

$m =$ 5

$$c = 5^{17} \bmod 33$$

$$c = 14$$

Letters are converted to numbers.
For the entered values the ciphertext is 14.

Step 4: Decrypt

For decryption the inverse formula is applied.
It uses the numbers of the private key:

$$m' = c^d \bmod n$$

Example:

$c =$ 14

$$m' = 14^{13} \bmod 33$$

$$m' = 5$$

The ASCII representation of the values you entered is:

► Sequence note

2. Faça a criptografia e descriptografia usando o algoritmo RSA:

- $p=3$, $q=11$, $e=7$, mensagem=5
- $p=5$, $q=11$, $e=3$, mensagem=9

Step 1: Set prime numbers

As a starting point for RSA, choose two prime numbers p and q :

$p =$ 5

$q =$ 11

For the algorithm to work, the two prime numbers must be different.

► Safety note

e (Step 2.3)

The number e can now be chosen freely.

However, it must be **coprime** to $\phi(n)$.

$e =$ 3

Step 3: Encrypt

To encrypt a number m to ciphertext c the following formula is applied.
It uses the numbers of the public key:

$$c = m^e \bmod n$$

RSA encrypts only numbers. These must be greater-equal 0 and less than $n = 55$.

Example:

$m =$ 9

$$c = 9^3 \bmod 55$$

$$c = 14$$

Step 4: Decrypt

For decryption the inverse formula is applied.
It uses the numbers of the private key:

$$m' = c^d \bmod n$$

Example:

$c =$ 14

$$m' = 14^{27} \bmod 55$$

$$m' = 9$$